

# Reflection and Traceability Report on Physiotherapy Companion

Team 11, technically functional

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## Contents

|          |                                                                                  |           |
|----------|----------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Changes in Response to Feedback</b>                                           | <b>3</b>  |
| 1.1      | SRS and Hazard Analysis . . . . .                                                | 3         |
| 1.2      | Design and Design Documentation . . . . .                                        | 5         |
| 1.3      | VnV Plan and Report . . . . .                                                    | 7         |
| <b>2</b> | <b>Extras</b>                                                                    | <b>10</b> |
| <b>3</b> | <b>Design Iteration (LO11 (PrototypeIterate))</b>                                | <b>11</b> |
| <b>4</b> | <b>Design Decisions (LO12)</b>                                                   | <b>11</b> |
| <b>5</b> | <b>Economic Considerations (LO23)</b>                                            | <b>11</b> |
| <b>6</b> | <b>Reflection on Project Management (LO24)</b>                                   | <b>13</b> |
| 6.1      | How Does Your Project Management Compare to Your Develop-<br>ment Plan . . . . . | 13        |
| 6.2      | What Went Well? . . . . .                                                        | 13        |
| 6.3      | What Went Wrong? . . . . .                                                       | 13        |
| 6.4      | What Would you Do Differently Next Time? . . . . .                               | 13        |
| <b>7</b> | <b>Ethics and Equity (LO18)</b>                                                  | <b>13</b> |
| <b>8</b> | <b>Reflection on Capstone</b>                                                    | <b>13</b> |
| 8.1      | Which Courses Were Relevant . . . . .                                            | 14        |
| 8.2      | Knowledge/Skills Outside of Courses . . . . .                                    | 14        |

[Reflection is an important component of getting the full benefits from a learning experience. Besides the intrinsic benefits of reflection, this document will be used to help the TAs grade how well your team responded to feedback. Therefore, traceability between Revision 0 and Revision 1 is an important part of the reflection exercise. In addition, several CEAB (Canadian Engineering Accreditation Board) Learning Outcomes (LOs) will be assessed based on your reflections. —TPLT]

# 1 Changes in Response to Feedback

## 1.1 SRS and Hazard Analysis

The hazard analysis document was improved through the peer review along with TA feedback in several areas for enhancement, including clarifying the roadmap with implementation timelines and rationales for safety requirements (#144), adding justification for future work prioritization (#150), and removing problematic assumptions about user familiarity with smart devices (#136). Inconsistencies between the SRS and Hazard Analysis were resolved (#129), while the system boundaries and components section was expanded with lower-level explanations to remove ambiguity (#135). The FMEA table was further decomposed to represent detailed failure modes rather than general effects (#142), and hazard categorization was clarified for technical hazards that could impact safety (#153). Additional improvements included adding rationales for each critical assumption (#136), addressing inconsistencies between assumptions and failure modes (#143), and noting where mitigation strategies would be expanded (#152). TA feedback further reinforced the need for roadmap rationales, which was addressed alongside similar peer feedback. One comment regarding bibliography completeness (#151) was determined to be invalid as the citation and reference were already properly included.

| Source                 | Issue # | Issue Description                                                                          | Changes Made                                                                                                                                     | Corresponding Commit |
|------------------------|---------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Peer Review            | #151    | Citation "IppolitoWallace1995" appears but bibliography incomplete                         | No change - comment invalid as bibliography and in-text citation were complete                                                                   | None                 |
| Peer Review (Group 14) | #144    | Unclear description of how/when safety requirements and hazard testing will be implemented | Added section in roadmap outlining rationale for accepting requirements within capstone timeline and rationales for excluded safety requirements | 939b502              |

| Source                                                    | Issue #              | Issue Description                                                                                                                            | Changes Made                                                                                             | Corresponding Commit            |
|-----------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------|
| Peer Review (Group 14)                                    | <a href="#">#150</a> | Safety requirements prioritized as "first" vs "in the future" without clear justification or timeline                                        | Added clear rationale for future work prioritization                                                     | 470F490                         |
| Peer Review (Group 14)                                    | <a href="#">#136</a> | Missing rationale for each assumption; assumption about user familiarity with smart device should be avoided                                 | Removed assumption about user familiarity; added rationales for each assumption                          | a103812                         |
| Peer Review (Group 14)                                    | <a href="#">#143</a> | Inconsistency between assumption about user familiarity and "confusing prompts" failure mode                                                 | Removed the initial assumption; will update language of "confusing prompts"                              | None (different commit)         |
| Peer Review (Group 14)                                    | <a href="#">#129</a> | Inconsistencies between SRS and Hazard Analysis; hazards need prioritization by severity                                                     | Updated consistency between SRS and HA                                                                   | Reference #410 PR               |
| Peer Review (Group 14)                                    | <a href="#">#135</a> | System components need lower-level explanation with sample hazards; authentication module lacks detail                                       | Added changes to HA system boundaries, splitting section to remove ambiguity                             | 4bd898a                         |
| Peer Review (Group 14)                                    | <a href="#">#142</a> | FMEA rows should represent detailed failure modes rather than effects                                                                        | Included further decomposition of table within FMEA                                                      | Different commit                |
| Peer Review (Group 14)                                    | <a href="#">#153</a> | Technical hazards could also be safety hazards leading to unsafe exercise                                                                    | Clarified hazard categorization                                                                          | 4381828                         |
| Peer Review (Group 14)                                    | <a href="#">#152</a> | Document lacks detailed mitigation strategies and implementation specifics                                                                   | Mitigation included; will add section about effectiveness                                                | None (different section)        |
| TA                                                        | N/A                  | For Section 8 (Roadmap), including rationale would be helpful for which safety requirements to implement and why they are in that time frame | Updated the section with timelines and rationale for why each safety requirement was placed in each term | Similar to <a href="#">#144</a> |
| Table 1: Feedback and changes for SRS and Hazard Analysis |                      |                                                                                                                                              |                                                                                                          |                                 |

## 1.2 Design and Design Documentation

In reference to changed made between Rev0 and Rev1, the MG was modified to reflect changes in the overall module design of the project. The MG modules were modified in their responsibilities and locations in the overall design (Refer to [Issue #408](#)). There were changes made to not only the modules themselves but the justification for anticipated changes, unlikely changes, and design decisions to further explain the effect they held on the final design of the project. It is hoped this additional level of explanation will help to clarify some of the limitations, assumptions, and constraints that were presented in several of the modules and other documents (Refer to [PR #416](#)). Changes to the modules required changes to the uses hierarchy and the traceability sections also found in the MG. Finally, there were modifications to the MG's timelien section to make it more focussed on modules and design documentation with specific dates rather than overall deliverables and general date ranges (Refer to [Issue #407](#)). Further changes are documented below in Table 2.

| Source                 | Issue #              | Issue Description                                                  | Changes Made                                                                                                                                   | Corresponding Commit |
|------------------------|----------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Peer Review (Group 14) | <a href="#">#242</a> | Hardware Hiding Module reference issues and Modules lacking detail | Hardware Hiding Module reference fixed, however module details did not seem to be lacking as per team and TA feedback; no change on this part. | 7edbcd4              |
| Peer Review (Group 14) | <a href="#">#241</a> | Module names unclear                                               | Module names updated for clarity. Home changed to Profile Activity, Main changed to Analysis Control Flow.                                     | d2b2239              |

| Source                 | Issue #              | Issue Description                                                                                                             | Changes Made                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Corresponding Commit |
|------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Peer Review (Group 14) | <a href="#">#244</a> | Home, Main, and Database modules seem multi-purposed. Consider rewording secrets or fracturing modules into several new ones. | The Main module was renamed to the Analysis Control Flow module and handles the overall logic for recording, so was unchanged. The Profile Activity module was updated to reflect the new responsibilities, then eventually scrapped as the UI was redesigned and absorbed the responsibilities. The Database module was decomposed into the Account Creation, Valid User, User Account, User Activity, and Exercise Data modules which helped to split secrets and make modules clearer. | adefa15              |
| TA                     | <a href="#">#407</a> | Timeline done at deliverable level rather than module implementation level and ambiguous team member assignments.             | Timeline updated to be more specifically focused on module implementation with specific target dates and team member assignments. Design documentation also included in the timeline, but other deliverables were removed.                                                                                                                                                                                                                                                                | bef9394              |
| TA                     | <a href="#">#408</a> | ACs, UCs, and Design Decisions lacking information.                                                                           | ACs, UCs, and Design Decisions updated with more rationale and impact on the overall design.                                                                                                                                                                                                                                                                                                                                                                                              | 8648a1c              |

Table 2: Feedback and changes for Module Guide

### 1.3 VnV Plan and Report

In terms of the Verification and Validation (V&V) Plan, the feedback was applied based on both the peer review and TA feedback. The main changes that have addressed include adding clear pass/fail criteria for test cases ([#167](#)), replacing vague script-based testing descriptions with detailed explanations of implemented unit tests ([#168](#), [#428](#), [#429](#)), and improving traceability between requirements and test cases ([#172](#)).

| Source      | Issue #              | Issue Description                                                                                                                      | Changes Made                                                                                                                                   | Commit  |
|-------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Peer Review | <a href="#">#167</a> | Add a clear pass/fail criteria                                                                                                         | Added clear pass/fail criteria                                                                                                                 | c8aa78e |
| Peer Review | <a href="#">#168</a> | Script-based testing lacks detail on how tests are performed                                                                           | Removed script-based testing and replaced with detailed descriptions of implemented unit test cases, documented individually in the V&V Report | c8aa78e |
| Peer Review | <a href="#">#169</a> | Fix revision date history                                                                                                              | Updated revision date history                                                                                                                  | cf936e1 |
| Peer Review | <a href="#">#170</a> | Symbols, abbreviations, and acronyms table lacks detail                                                                                | Expanded and improved table detail                                                                                                             | c8aa78e |
| Peer Review | <a href="#">#171</a> | The document references external LaTeX input files on lines 18-19 that may not exist or be properly linked in the repository structure | No change made                                                                                                                                 | None    |

|             |                      |                                                                            |                                                                                                                                                                            |         |
|-------------|----------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Peer Review | <a href="#">#172</a> | Lack of traceability between requirements and unit tests                   | Revised to include the correct traceability mapping between both                                                                                                           | c8aa78e |
| Peer Review | <a href="#">#173</a> | Unclear distinction between system testing and functional testing sections | Updated the respective sections and added introductions to specify what will be covered in each section                                                                    | 8912684 |
| Peer Review | <a href="#">#174</a> | Explanations for relevance of documentation are vague                      | Increased level of detail in explanations                                                                                                                                  | c8aa78e |
| Peer Review | <a href="#">#175</a> | Lack of detail in the Verification and Validation Plan section             | Expanded and clarified section content                                                                                                                                     | c8aa78e |
| TA Feedback | <a href="#">#414</a> | Table formatting could be improved                                         | Cleaned and improved table formatting                                                                                                                                      | c8aa78e |
| TA Feedback | <a href="#">#427</a> | Missing detail in automated testing and implementation verification        | Added more specific detail to these sections                                                                                                                               | c8aa78e |
| TA Feedback | <a href="#">#428</a> | Script-based testing descriptions lack clarity on execution                | Replaced script-based testing with detailed descriptions of all of the implemented unit test cases. Each of these test cases are documented individually in the V&V Report | c8aa78e |



|             |                      |                                                              |                                                                                                                                                                            |         |
|-------------|----------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| TA Feedback | <a href="#">#429</a> | Ambiguity in scripting for functional testing                | Replaced script-based testing with detailed descriptions of all of the implemented unit test cases. Each of these test cases are documented individually in the V&V Report | c8aa78e |
| TA Feedback | <a href="#">#430</a> | Missing detail in code walkthroughs and static testing       | Added additional detail to these sections                                                                                                                                  | c8aa78e |
| TA Feedback | <a href="#">#431</a> | Missing reference in-text citation in the Objectives section | Added the missing reference                                                                                                                                                | c8aa78e |

Table 3: Feedback and changes for Verification and Validation Plan

In terms of the Verification and Validation (V&V) Report, the feedback was applied based on the TA feedback in several sections. There were no peer review issues found for this document, so we based it solely on the TA feedback provided through Avenue. Some of the changes that we applied included improving the level of detail when comparing the expected vs. actual result of the outputs ([#432](#)), fixing the table formatting for section 7 ([#433](#)), and ensuring the traceability between the V&V Plan and V&V Report aligned and matched up with one another ([#434](#)).

| Source | Issue # | Issue Description | Changes Made | Commit |
|--------|---------|-------------------|--------------|--------|
|--------|---------|-------------------|--------------|--------|

|             |                      |                                                                                                                                                                                       |                                                                                                                                                                     |         |
|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| TA Feedback | <a href="#">#432</a> | The document would benefit greatly from having additional comparisons between expected output and actual output to put the results in a more meaningful context                       | The actual result column has been updated to have a more meaningful comparison to the expected output result and provide more detailed context of the actual result | 4353716 |
| TA Feedback | <a href="#">#433</a> | The tables in section 7 all need to be fixed; the indentation, justification and alignment when compared to the rest of the document makes it seem unprofessional and should be fixed | Fixed formatting of the tables                                                                                                                                      | 4353716 |
| TA Feedback | <a href="#">#434</a> | Traceability between the V&V Plan and V&V Report is inconsistent                                                                                                                      | Improved traceability between the two documents to ensure alignment and consistency across test cases                                                               | e5f3a37 |

Table 4: Feedback and changes for Verification and Validation Report

## 2 Extras

[Summarize the extras that were tackled by this project. Extras can include usability testing, code walkthroughs, user documentation, formal proof, Gender-Mag personas, Design Thinking, etc. Extras should have already been approved by the course instructor as included in your problem statement. —TPLT]

### 3 Design Iteration (LO11 (PrototypeIterate))

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don't just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

### 4 Design Decisions (LO12)

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

### 5 Economic Considerations (LO23)

Yes, there is a clear market for our product. Existing physiotherapy companion apps such as [Physiapp](#) and [Physiomobile](#) offer progress tracking (checklist-based) and gamification (streaks), but they lack real-time exercise form feedback. Our app fills this gap by using OpenCV and MediaPipe pose landmarks to analyze patient movement during home exercises, giving immediate corrective feedback. Additionally, we provide value back to the physiotherapist by automatically sharing tracked metrics (e.g., reps, form quality) together with subjective patient data (pain levels, perceived exertion). This addresses the gap for patients and physios between appointments.

#### Marketing Approach

Marketing would target physiotherapy clinics and independent practitioners. Strategies include:

- Direct outreach to private clinics via email and informational webinars.
- Free trial periods for clinics to demonstrate improved patient adherence and outcomes.
- Testimonials and case studies from early-adopter clinics.

#### Cost to Produce Application

We estimate the cost to develop our application to a production-ready version (iOS/Android app, backend dashboard for physios, pose-analysis pipeline) at:

- Development (4 months, 2-3 engineers): \$60,000 - \$80,000.
- Exercise pose analysis expertise (consultation with physiotherapists to validate exercise models and feedback logic): \$10,000 - \$15,000.

- Cloud infrastructure (AWS/GCP for pose inference): \$5,000 initial setup + \$500/month.
- Legal/regulatory (privacy, medical device classification, liability for recording and providing feedback): \$15,000 - \$20,000
- Marketing launch: \$5,000.

**Total initial cost: approximately \$95,000 - \$125,000.**

## Pricing Model

We would not charge individual patients. Instead, we adopt a subscription model for physiotherapy clinics, tiered by the number of active patients per month. A “small private clinic” typically has 2-3 physiotherapists, each seeing 25-50 patients per week. Therefore, reasonable patient tiers are:

- **Tier 1 (up to 50 active patients / month):** \$50 / month
- **Tier 2 (up to 150 active patients / month):** \$100 / month
- **Tier 3 (unlimited patients):** \$200 / month

## Breakeven Analysis

Assume an average subscription of \$100 / month per clinic (weighted toward Tier 2). Monthly recurring costs (cloud, support) are roughly \$500 for every 100 clinics. Development cost is a one-time \$110,000.

Monthly profit per clinic after variable costs: \$100 - \$5 = \$95. Number of clinic-months needed to recover development cost:

$$\frac{\$110,000}{\$95 \text{ per clinic-month}} \approx 1,158 \text{ clinic-months.}$$

If we acquire 100 clinics in the first year, breakeven occurs after about 11.6 months. In terms of total clinics, we would need roughly 97 clinics subscribing for one full year to break even.

## Potential Users (TAM / SOM)

- **Total Addressable Market (TAM):** All physiotherapists in Canada - approximately 22,000 registered physiotherapists (Canadian Institute for Health Information). Assuming an average of 5 physiotherapists per private clinic (typical for a medium-sized clinic), that gives roughly 4,400 clinics.
- **Serviceable Available Market (SAM):** Private clinics in Ontario, where our initial launch would focus. Ontario has roughly 6,000 physiotherapists, translating to about 1,200 private clinics.

- **Serviceable Obtainable Market (SOM):** In the first two years, with targeted marketing, we could realistically capture 1% of the Ontario market - about 12 clinics. That would generate monthly revenue of \$1,200 (at \$100 average). With successful promotion and marketing of the application, the application could grow to 5% (60 clinics) after three years, achieving monthly revenue of \$6,000 and approaching a breakeven point.

## 6 Reflection on Project Management (LO24)

[This question focuses on processes and tools used for project management. —TPLT]

### 6.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use the technology you planned on using? —TPLT]

### 6.2 What Went Well?

[What went well for your project management in terms of processes and technology? —TPLT]

### 6.3 What Went Wrong?

[What went wrong in terms of processes and technology? —TPLT]

### 6.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

## 7 Ethics and Equity (LO18)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

## 8 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

## **8.1 Which Courses Were Relevant**

[Which of the courses you have taken were relevant for the capstone project?  
—TPLT]

## **8.2 Knowledge/Skills Outside of Courses**

[What skills/knowledge did you need to acquire for your capstone project that  
was outside of the courses you took? —TPLT]